Presentation notes

Literature search · How many weeks out do you look? · ~3 min

1 · Title

My paper estimates how many weeks out from a show a Classic Physique competitor *looks*, from photos, using pairwise ranking. This assignment is the literature search behind it.

2 · One question, three angles

The search question: how has progress toward contest leanness, or body composition in general, been estimated from photos? It splits into three parts:

- **Q1:** body fat from images
- **Q2:** body-composition tracking during contest prep
- **Q3:** methods that learn an order or progression from images

A fourth block covers AI applied to bodybuilding.

3 · Method

- Started in Google Scholar Labs to collect vocabulary and a handful of key seed papers.
- Web of Science wouldn’t log in and Scopus access is pending, so the searches ran in **OpenAlex**. It covers what Scholar finds, but can filter by year, language and type, and export results.
- Same filters on every query: 2016–2025, English, articles, reviews, conference papers and book chapters.

4 · Queries

- Four final queries, one per sub-question plus AI in bodybuilding: **1,251** unique records.
- First run had no conference papers, where most ML work is published, so I added them.
- Body-fat query missed key papers, so I widened it. Removing that part entirely gave 3,991 hits, too many.
- Ranking query, four versions *(point at chart)*:
  - Original, **292**: matched “Siam”/Thailand and cancer progression
  - Neural-network term required, **46**: too few
  - Image term removed, **703**: mostly irrelevant
  - Final, image term required: **138**

5 · Results

- Cleaning removed **84**: spam ads, retractions, abstract books, preprints, non-English titles, duplicates.
- Added **4** key seed papers by hand that the queries missed.
- **Dataset 1:** 1,171 papers, under the 1,500 limit.
- **Dataset 2:** 50 screened papers, each with a theme and reason:
  - 15 body fat from photos
  - 13 contest prep
  - 11 ranking methods
  - 9 shape or weight from images
  - 2 AI in bodybuilding

6 · Gap and next steps

Body fat from photos, contest-prep body composition and image ranking are each well studied, but none of the 50 papers combines them to judge contest readiness. That’s the gap my paper targets.

Next steps

1. Mind-map the 50 papers by theme; talk the idea through with 2–3 colleagues.
2. Next assignment: write the idea as an introduction in LaTeX (KOMA-Script).
3. TikZ figure: two photos → pairwise ranker → apparent weeks out.
4. 3–4 research objectives and highlights.
5. When Scopus access arrives: re-run the same four queries, merge by DOI.

Likely questions

Why not Scholar directly?

No filters for document type or language, and no export.

Why is dataset 1 noisy?

It’s the raw search after cleaning, deliberately not screened. The screening is in dataset 2.

Years off by one?

Online-first publication years, so a few papers show the year before their print issue.

Why add papers by hand?

Known key papers whose wording fell outside the query terms. Standard snowballing; marked in the found_by column.

Exact wording of b3v2?

Not saved. It was the original ranking query with a neural-network term made mandatory: 46 papers.